

Why Do Uber Rides Get Cancelled? A Data-Driven Approach

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Data Overview

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Uber Data Analytics 2024

This dataset provides detailed Uber ride data for the year 2024, enabling in-depth analysis of booking trends, vehicle performance, revenue generation, cancellation patterns, and customer satisfaction.

The dataset includes 148,770 total bookings across multiple vehicle types, offering a comprehensive view of ride-sharing operations. It captures key aspects such as completed rides, cancellations, customer behavior, revenue metrics, and overall service performance.



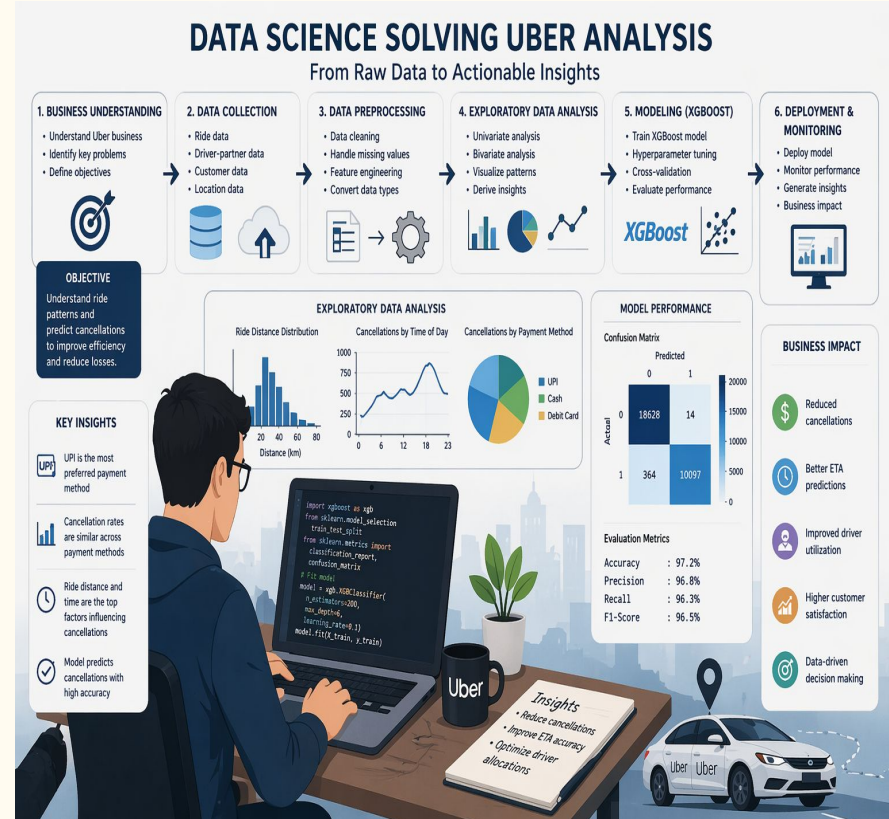
GOAL

What real-world problem we are trying to solve?

In on-demand platforms (ride-sharing, delivery, logistics), cancellations create inefficiencies, increase costs, and negatively impact user satisfaction.

Therefore, our goal is...

To build a predictive model for ride cancellations and analyze the most important factors affecting customer and driver cancellation behavior.



Features Explained (21)

Some of the key features:

Booking Status – Status of the ride (Completed, Cancelled by Customer, Cancelled by Driver, etc.)

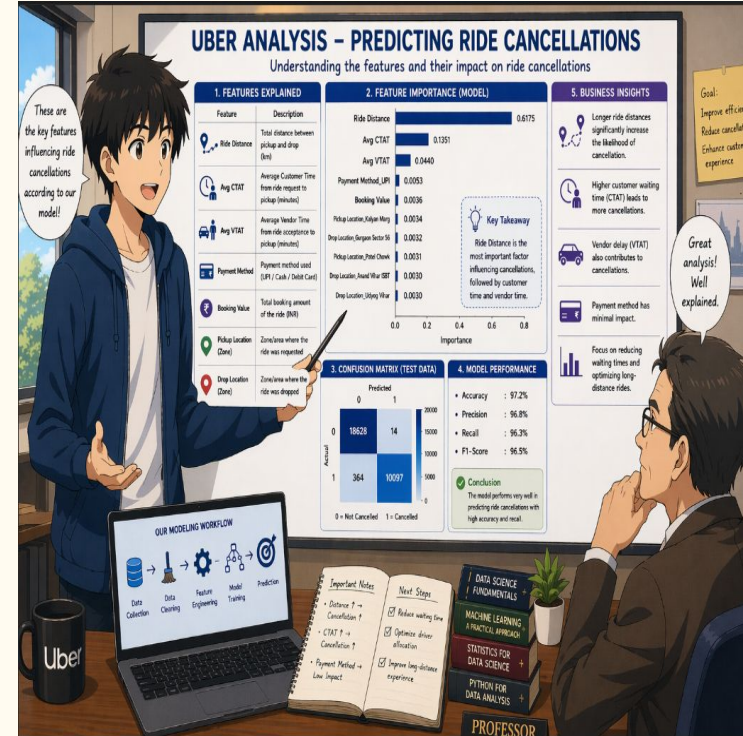
Vehicle Type – Type of vehicle (Go Mini, Go Sedan, Auto, Bike, UberXL, Premier Sedan)

Pickup Location – Starting point of the ride

AVG VTAT - Average time taken by the driver to reach the pickup location (minutes)

AVG CTAT - Average trip duration

Payment Type - Mode of Payment



```
[5]: df = pd.read_excel(r"C:\Users\tsrap\Downloads\ncr_ride_bookings.csv.xlsx")
```

```
[6]: df.head()
```

```
[6]:
```

	Date	Time	Booking ID	Booking Status	Customer ID	Vehicle Type	Pickup Location	Drop Location	Avg VTAT	Avg CTAT	...	Reason for cancelling by Customer	Cancelled Rides by Driver	Driver Cancellation Reason	Incomplete Rides
0	2024-03-23	12:29:38	"CNR5884300"	No Driver Found	"CID1982111"	eBike	Palam Vihar	Jhilmil	NaN	NaN	...	NaN	NaN	NaN	NaN
1	2024-11-29	18:01:39	"CNR1326809"	Incomplete	"CID4604802"	Go Sedan	Shastri Nagar	Gurgaon Sector 56	4.9	14.0	...	NaN	NaN	NaN	1.0
2	2024-08-23	08:56:10	"CNR8494506"	Completed	"CID9202816"	Auto	Khandsa	Malviya Nagar	13.4	25.8	...	NaN	NaN	NaN	NaN
3	2024-10-21	17:17:25	"CNR8906825"	Completed	"CID2610914"	Premier Sedan	Central Secretariat	Inderlok	13.1	28.5	...	NaN	NaN	NaN	NaN
4	2024-09-16	22:08:00	"CNR1950162"	Completed	"CID9933542"	Bike	Ghitorni Village	Khan Market	5.3	19.6	...	NaN	NaN	NaN	NaN

5 rows × 21 columns

Data Cleaning

- **Converted** columns to appropriate **data types** (e.g., date, time, numeric) to ensure accurate analysis
- **Checked for missing/null** values across all columns
- **Created a new target variable** `is_cancelled` for modeling purposes

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150000 entries, 0 to 149999
```

```
Data columns (total 21 columns):
```

#	Column	Non-Null Count	Dtype
0	Date	150000 non-null	datetime64[ns]
1	Time	150000 non-null	object
2	Booking ID	150000 non-null	object
3	Booking Status	150000 non-null	object
4	Customer ID	150000 non-null	object
5	Vehicle Type	150000 non-null	object
6	Pickup Location	150000 non-null	object
7	Drop Location	150000 non-null	object
8	Avg VTAT	139500 non-null	float64
9	Avg CTAT	102000 non-null	float64
10	Cancelled Rides by Customer	10500 non-null	float64
11	Reason for cancelling by Customer	10500 non-null	object
12	Cancelled Rides by Driver	27000 non-null	float64
13	Driver Cancellation Reason	27000 non-null	object
14	Incomplete Rides	9000 non-null	float64
15	Incomplete Rides Reason	9000 non-null	object
16	Booking Value	102000 non-null	float64
17	Ride Distance	102000 non-null	float64
18	Driver Ratings	93000 non-null	float64
19	Customer Rating	93000 non-null	float64
20	Payment Method	102000 non-null	object

```
dtypes: datetime64[ns](1), float64(9), object(11)
```

```
memory usage: 24.0+ MB
```

The columns “Incomplete Rides”, “Cancelled Rides by Driver”, and “Cancelled Rides by Customer” contain only two values:

- 1 → Indicates the event occurred (e.g., ride cancelled or incomplete)
- NaN → Indicates the event did not occur or data is missing

This shows that these columns act as binary flags, where missing values effectively represent “No”.

Defining the Target Variable

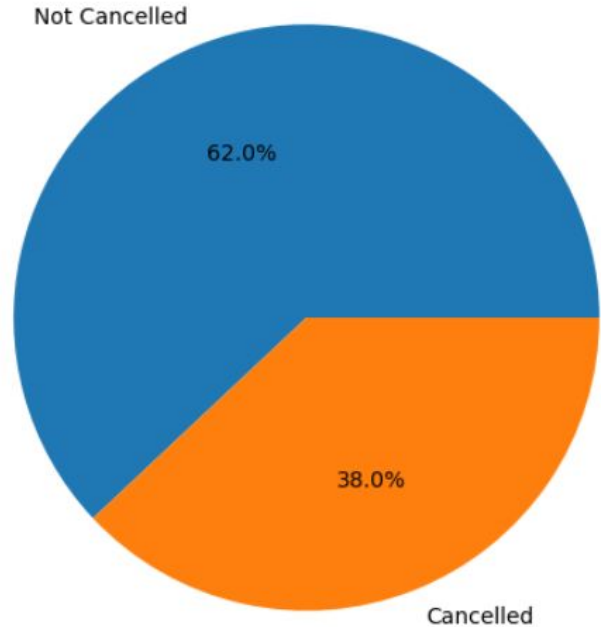
Define Target Variable

```
[19]: df['Booking Status'].unique()
```

```
[19]: array(['No Driver Found', 'Incomplete', 'Completed',  
        'Cancelled by Driver', 'Cancelled by Customer'], dtype=object)
```

```
[20]: df['is_cancelled'] = df['Booking Status'].isin([  
        'Cancelled by Driver',  
        'Cancelled by Customer',  
        'No Driver Found',  
        'Incomplete'  
    ]).astype(int)
```

Cancellation Distribution



Exploratory Data Analysis

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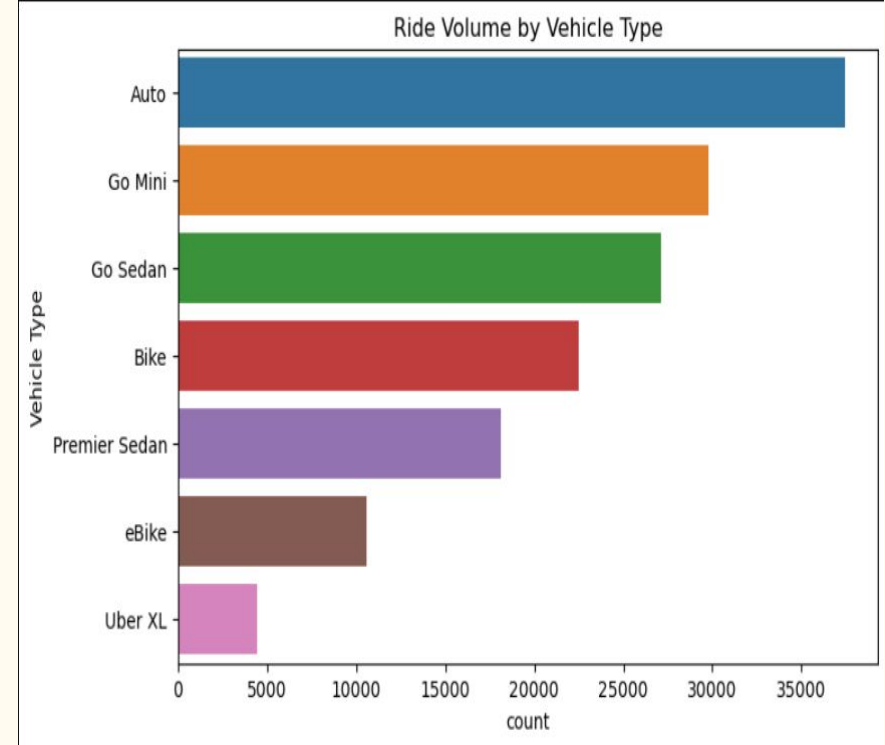
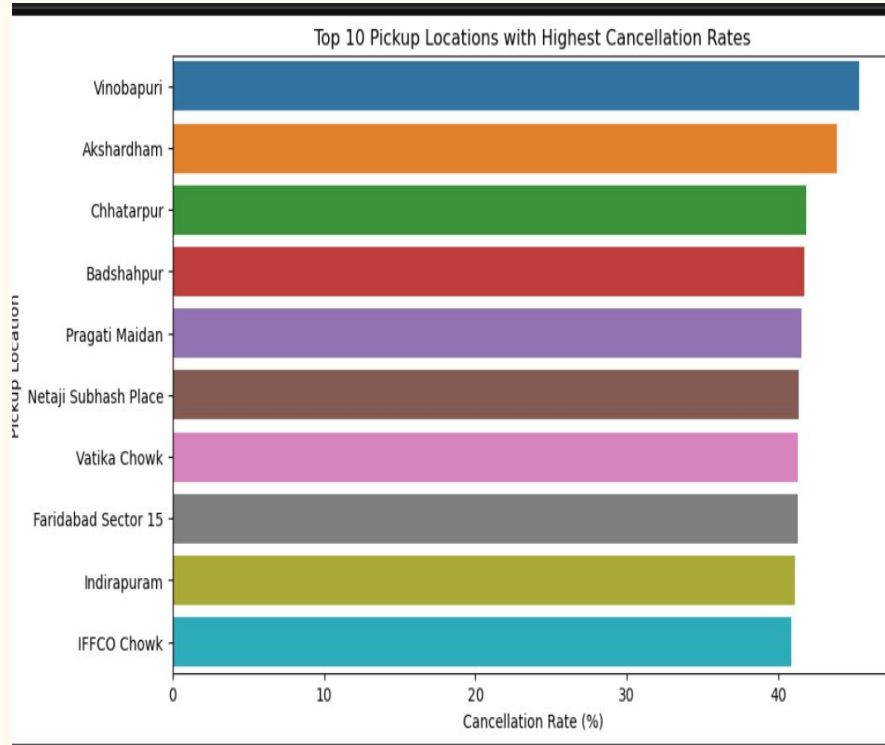
Analyze Cancellation reasons by the Driver

```
: df['Reason for cancelling by Customer'].value_counts()  
df['Driver Cancellation Reason'].value_counts()
```

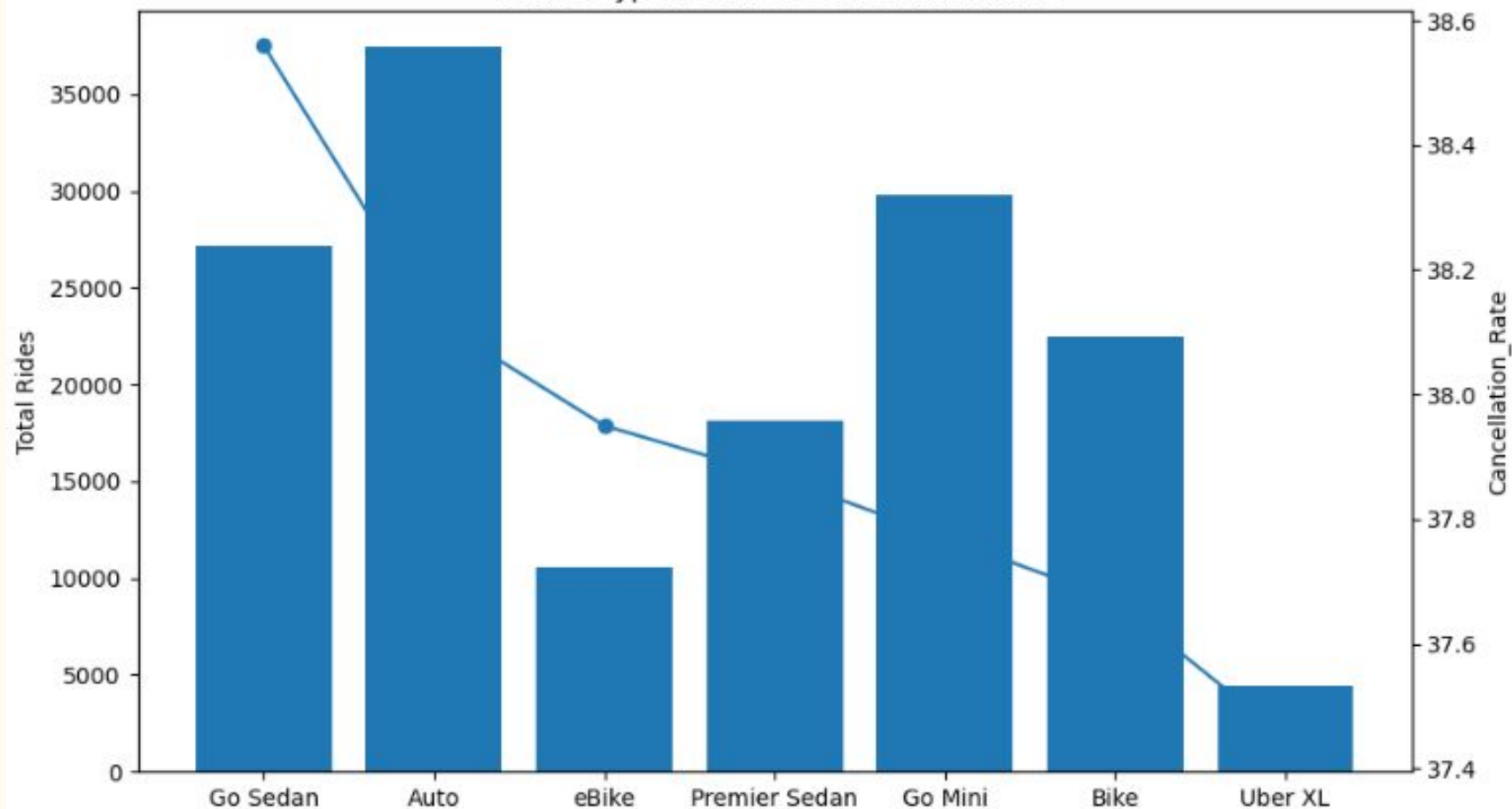
```
: Driver Cancellation Reason  
Customer related issue          6837  
The customer was coughing/sick   6751  
Personal & Car related issues    6726  
More than permitted people in there 6686  
Name: count, dtype: int64
```

- Shows the most common reasons why drivers cancel trips
- Helps identify the biggest pain points
- Allows Uber to fix issues and reduce cancellations

Analyze Cancellation Rate Based on Location and Vehicle Type wrt Target variable



Vehicle Type: Count vs Cancellation Rate



Time Based Patterns

```
def get_time_period(hour):  
    if 5 <= hour < 9:  
        return 'Early Morning'  
    elif 9 <= hour < 12:  
        return 'Morning'  
    elif 12 <= hour < 17:  
        return 'Afternoon'  
    elif 17 <= hour < 21:  
        return 'Evening'  
    else:  
        return 'Night'  
  
df['time_period'] = df['hour'].apply(get_time_period)
```

```
df.groupby('time_period')['is_cancelled'].mean().mul(100)
```

```
time_period  
Afternoon      38.270580  
Early Morning  38.012151  
Evening        38.165828  
Morning        37.708484  
Night          37.566069  
Name: is_cancelled, dtype: float64
```

- Shows when cancellations are higher, helping Uber adjust driver availability and pricing to reduce cancellations.
- For example: If more cancellations happen in the evening, Uber can raise prices or give incentives so more drivers come, rides get assigned faster, and fewer people cancel.

Idea is...

Higher price → more drivers come online → faster matches → fewer cancellations

How Time Factors Influence Ride Cancellations

- Avg CTAT and Avg VTAT have a very weak (almost negligible) positive correlation (~ 0.06), meaning they are essentially not related.
- Customer cancellations show a positive relation with Avg VTAT (~ 0.31) $\rightarrow$ higher driver delay increases customer cancellations

```
: df['customer_cancelled'] = df['Cancelled Rides by Customer'].notnull().astype(int)
df['driver_cancelled'] = df['Cancelled Rides by Driver'].notnull().astype(int)

: cols = ['Avg CTAT', 'Avg VTAT', 'customer_cancelled', 'driver_cancelled']

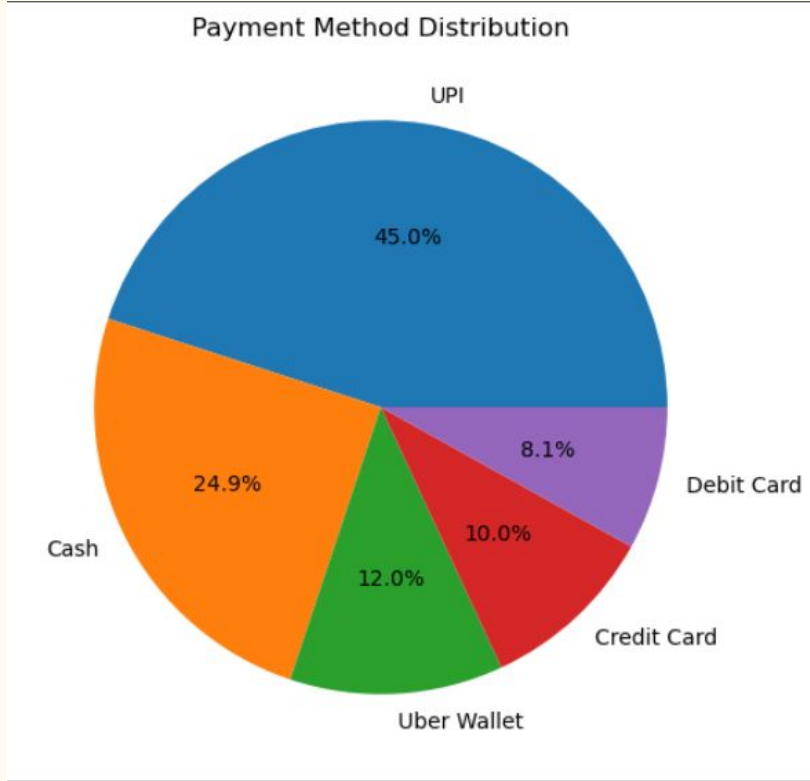
corr = df[cols].corr()
print(corr)
```

	Avg CTAT	Avg VTAT	customer_cancelled	driver_cancelled
Avg CTAT	1.000000	0.062175	NaN	NaN
Avg VTAT	0.062175	1.000000	0.306655	-0.123904
customer_cancelled	NaN	0.306655	1.000000	-0.128540
driver_cancelled	NaN	-0.123904	-0.128540	1.000000

```
: df.groupby('driver_cancelled')[['Avg CTAT', 'Avg VTAT']].mean()
df.groupby('customer_cancelled')[['Avg CTAT', 'Avg VTAT']].mean()
```

	Avg CTAT	Avg VTAT
customer_cancelled		
0	29.149636	8.126210
1	NaN	12.512381

Payment method Impact



```
df.groupby('Payment Method')['is_cancelled'].mean()

Payment Method
Cash          0.088816
Credit Card   0.087080
Debit Card     0.086540
UPI            0.088763
Uber Wallet    0.087162
Name: is_cancelled, dtype: float64
```

- UPI is the most preferred payment method
- Cancellation rates are similar across all methods, suggesting payment type has little impact on cancellations.

Data Cleaning

Goal: Predict cancellation before it happens.

But these columns ('Cancelled Rides by Customer', 'Reason for cancelling by Customer', 'Cancelled Rides by Driver', 'Driver Cancellation Reason', 'Incomplete Rides', 'Incomplete Rides Reason') tells you that cancellation already happened.

This could cause data leakage problem!!

```
df.isnull().mean().sort_values(ascending=False)
```

Incomplete Rides	0.94
Incomplete Rides Reason	0.94
Reason for cancelling by Customer	0.93
Cancelled Rides by Customer	0.93
Driver Cancellation Reason	0.82
Cancelled Rides by Driver	0.82
Customer Rating	0.38
Driver Ratings	0.38
Payment Method	0.32
Booking Value	0.32
Ride Distance	0.32
Avg CTAT	0.32
Avg VTAT	0.07
is_cancelled	0.00
hour	0.00
time_period	0.00
day_of_week	0.00
day_name	0.00
customer_cancelled	0.00
Date	0.00
Time	0.00
Drop Location	0.00
Pickup Location	0.00
Vehicle Type	0.00
Customer ID	0.00
Booking Status	0.00
Booking ID	0.00
driver_cancelled	0.00
dtype: float64	

Treating Missing Values

- 1) We dropped the columns that has $>90\%$ of missing values
- 2) We treated 'Booking Value', 'Ride Distance', 'Avg CTAT', 'Avg VTAT' with median and Payment method with mode
- 3) Ratings are given AFTER the ride is completed. So, we dropped it.
- 4) We dropped 'customer_cancelled' and 'driver_cancelled' as well because we already have the target variable

```
df_model.shape
(150000, 15)

df_model.isnull().mean().sort_values(ascending=False)
```

Date	0.0
Time	0.0
Vehicle Type	0.0
Pickup Location	0.0
Drop Location	0.0
Avg VTAT	0.0
Avg CTAT	0.0
Booking Value	0.0
Ride Distance	0.0
Payment Method	0.0
is_cancelled	0.0
hour	0.0
time_period	0.0
day_of_week	0.0
day_name	0.0
dtype:	float64

Model Building

—

```
models = []
models.append(('Logistic Regression', LogisticRegression(random_state=42)))
models.append(('Naive bayes', GaussianNB()))
models.append(('KNN', KNeighborsClassifier()))
models.append(('Random Forest', RandomForestClassifier(random_state = 42)))
models.append(('SGD Classifier', SGDClassifier(random_state=42)))
models.append(('XGBoost', XGBClassifier(random_state = 42)))
models.append(('GradientBoosting', GradientBoostingClassifier(random_state = 42)))
results = []
names = []
```

Top Models...

1. Random Forest and Gradient Boosting
2. XGBoost
3. KNN
4. Logistic Regression

```
for name, model in models:
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print("{} : {}".format(name, accuracy))
```

C:\Users\tsrap\anaconda3\Lib\site-packages\sklearn\STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

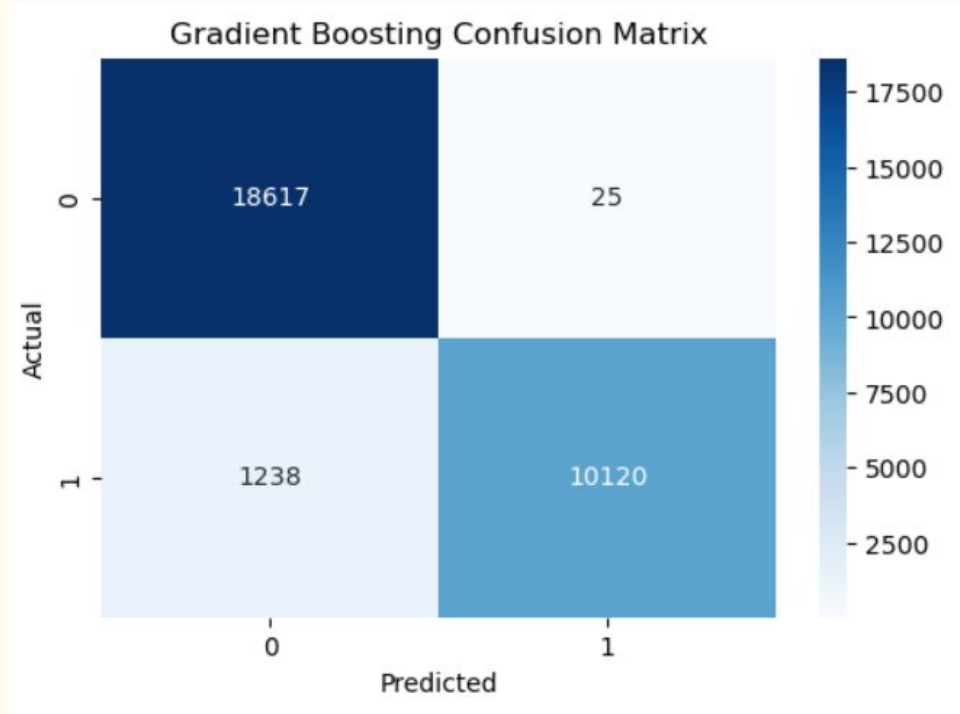
Increase the number of iterations (max_iter) or scale the data (see the documentation for alternative methods).
Please also refer to the documentation for alternative methods.
<https://scikit-learn.org/stable/modules/preproc.html>
https://scikit-learn.org/stable/modules/linear_model.html
n_iter_i = _check_optimize_result(

```
Logistic Regression : 0.8102333333333334
Naive bayes : 0.7041
KNN : 0.9439
Random Forest : 0.9579
SGD Classifier : 0.6746333333333333
XGBoost : 0.9558
GradientBoosting : 0.9579333333333333
```

Gradient Boosting

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Gradient Boosting - Confusion Matrix



- Model correctly predicts a significant amount of non-cancelled rides with the True negative being 18617
- Minimal false positives (25)
- Has a high accuracy and model performance

Gradient Boosting - Feature Importance

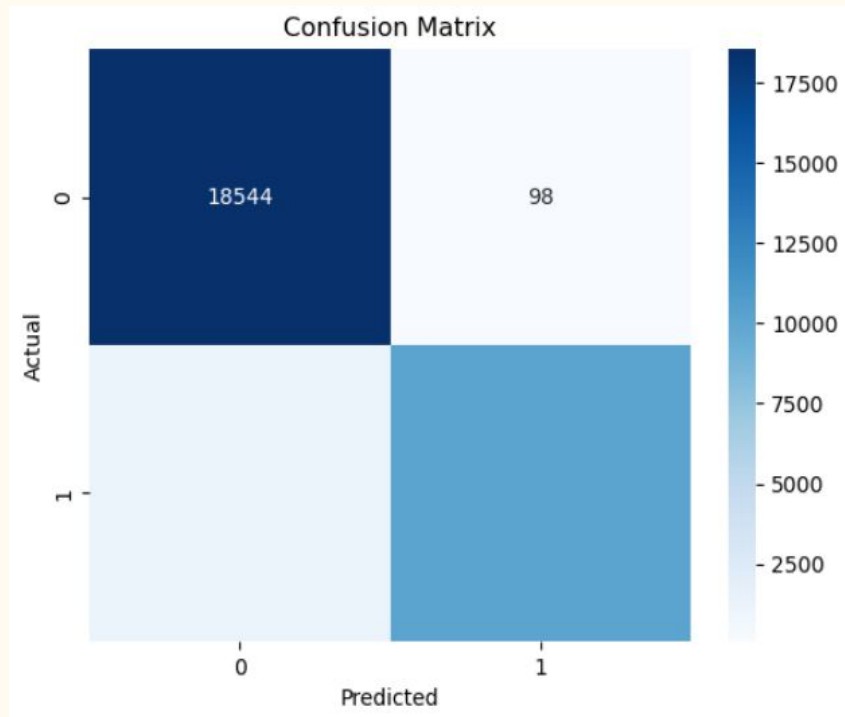
```
01. Ride Distance: 0.511577
02. Avg CTAT: 0.469629
03. Avg VTAT: 0.016697
04. Booking Value: 0.001305
05. Payment Method_UPI: 0.000407
06. Pickup Location_Tilak Nagar: 0.000029
07. Pickup Location_Seelampur: 0.000017
08. Pickup Location_Jasola: 0.000016
09. Drop Location_Delhi Gate: 0.000015
10. Drop Location Tilak Nagar: 0.000014
```

- **Ride distance** is the most important feature being over half the importance score
- **Average Customer time** makes up a majority of the rest of the importance score with average vendor time being third
- The **distance and sequential the time** are the most key factors here

XGBoosting

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XGBoost – Confusion Matrix



- Model correctly predicts a large number of non-cancelled rides (True Negatives: 18628)
- Very few false positives (14) → rarely predicts cancellation incorrectly
- Good performance in identifying both cancelled and non-cancelled rides

Overall indicates high accuracy and strong model performance

XGBoost – Feature Importance

```
importance = pd.DataFrame({  
    'Feature': X_train.columns,  
    'Importance': xgb.feature_importances_  
}).sort_values(by='Importance', ascending=False)  
  
print(importance.head(10))
```

	Feature	Importance
3	Ride Distance	0.603652
1	Avg CTAT	0.131796
0	Avg VTAT	0.045019
364	Payment Method_UPI	0.010655
97	Pickup Location_Lok Kalyan Marg	0.003451
233	Drop Location_Gurgaon Sector 56	0.003102
2	Booking Value	0.003056
22	Pickup Location_Aya Nagar	0.002801
92	Pickup Location_Kherki Daula Toll	0.002764
100	Pickup Location_Maidan Garhi	0.002756

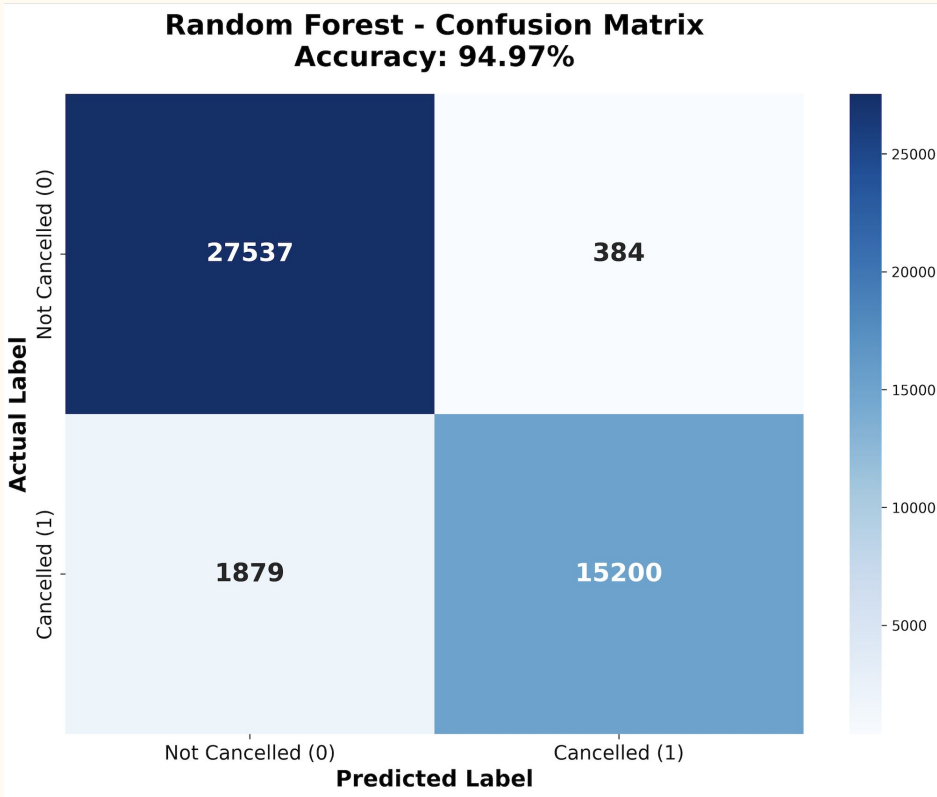
- **Ride Distance** is the most important feature (~61.7%) → strongest predictor of cancellations
- **Avg CTAT** (Customer Time) is the second most important (~13.5%)
- **Avg VTAT** (Vendor Time) also contributes but less (~4.4%)
- Other features like Payment Method (UPI) and Booking Value have very low impact
- Location-based features (pickup/drop) have minimal influence

Insight: Trip-related factors (distance & time) matter more than payment type or location

Random Forest

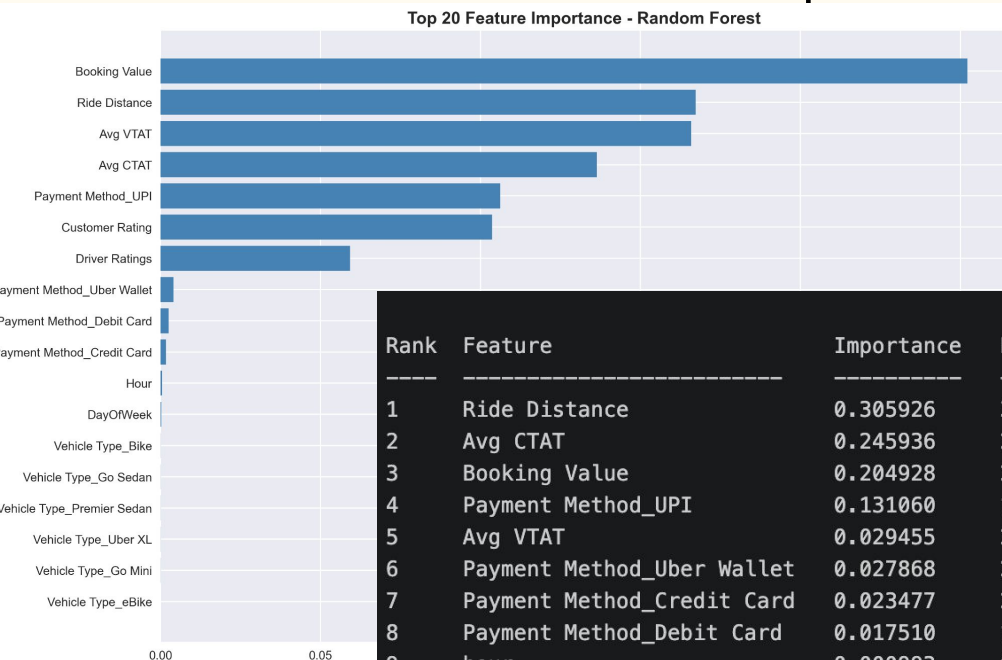
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Random Forest: Confusion Matrix



- The model correctly predicts a very large number of non-cancelled rides (**True Negatives: 27,537**), showing strong performance on the majority class
- It also successfully identifies a high number of cancelled rides (**True Positives: 15,200**), indicating effective detection of cancellations
- The number of false positives is very low (**384**), meaning the model rarely flags normal rides as cancelled
- Some cancellations are missed (**False Negatives: 1,879**), but this is relatively small compared to correctly identified cases
- The model achieves a high overall accuracy of **94.97%**, reflecting strong predictive capability
- Overall, the model maintains a good balance between **detecting cancellations** and **avoiding unnecessary false alarms**

Random Forest: Feature Importances



Rank	Feature	Importance	Percentage
1	Ride Distance	0.305926	30.59%
2	Avg CTAT	0.245936	24.59%
3	Booking Value	0.204928	20.49%
4	Payment Method_UPI	0.131060	13.11%
5	Avg VTAT	0.029455	2.95%
6	Payment Method_Uber Wallet	0.027868	2.79%
7	Payment Method_Credit Card	0.023477	2.35%
8	Payment Method_Debit Card	0.017510	1.75%
9	hour	0.000993	0.10%
10	day_name_Sunday	0.000308	0.03%
11	day_name_Saturday	0.000264	0.03%
12	day_name_Monday	0.000237	0.02%
13	day_name_Tuesday	0.000210	0.02%
14	Vehicle Type_Go Mini	0.000209	0.02%
15	day_name_Wednesday	0.000208	0.02%

- **Ride Distance (30.59%)** is the most influential feature, indicating that longer trips significantly increase the likelihood of cancellations, likely due to higher time and effort commitment
- **Avg CTAT (24.59%)** is the second most important factor, showing that total travel time plays a major role in user decisions, especially when delays are involved
- **Booking Value (20.49%)** also has strong impact, suggesting that higher-cost rides are more prone to cancellations due to financial considerations
- **Payment Method – UPI (13.11%)** contributes meaningfully, indicating that payment preferences may reflect user behavior patterns related to cancellations
- **Avg VTAT (2.95%)** has a smaller but noticeable effect, reinforcing that wait time still influences decisions, though less than total travel time
- Other payment methods (wallet, credit, debit) contribute **minor influence (~1–3%)**, suggesting limited impact on predictions
- Features like **hour of day, day of week, and vehicle type (<0.1%)** have negligible importance, indicating they do not significantly affect cancellation behavior

Logistic Regression

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Model Evaluation

Target Variable: “is_cancelled” - Did the ride **Fail (1)** or **Succeed (0)**

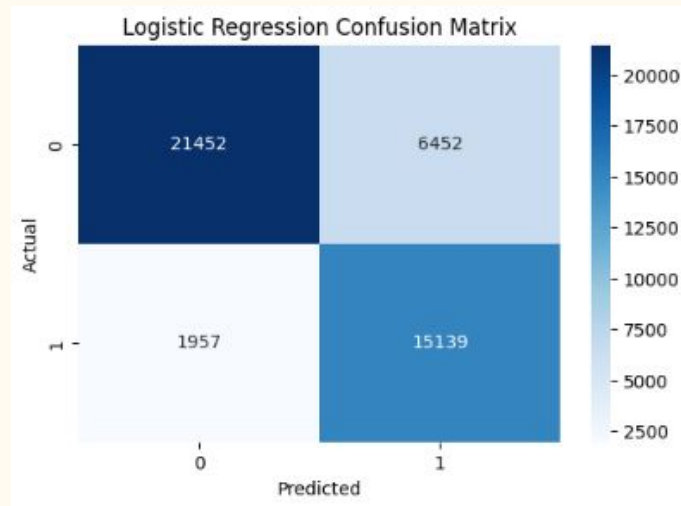
Model Performance:

- Accuracy: **81%**
- Recall- Correctly **Identifies** (Cancellations): **89%**
- Precision (Cancellations): **70%**

Results:

- Model is **effective** at **identifying** cancellations
- Prioritizes **recall** over **precision**
- Fewer missed cancellations (**FN = 1,957**)
- **Trade-off:** More false positives (FP = 6,452) but acceptable because missing cancellations is worse than false alarms

Confusion Matrix & Classification Report



Classification Report				
	precision	recall	f1-score	support
0	0.92	0.77	0.84	27984
1	0.70	0.89	0.78	17096
accuracy			0.81	45080
macro avg	0.81	0.83	0.81	45080
weighted avg	0.83	0.81	0.82	45080

Model Insights

1. What **factors** most **influence cancellations**?
2. How does **payment method** affect cancellations?
3. Do **ride characteristics** (distance, value) affect cancellations?

1. **Payment Method** (strongest factor)
 - a. Ride distance & trip duration
 - b. Pickup/drop location
 - c. Weekend days (Saturday & Sunday)
2. **Digital Payments** → **lower cancellation rates**
 - a. Cash → slightly higher cancellations
 - b. Users are more committed when payment is pre-set
3. **Longer Rides** → less likely to cancel
 - a. Higher booking value → lower cancellations
 - b. Indicates stronger commitment for higher-value rides

Top Features Increasing Cancellation (Positive Coefficients):		
	Feature	Coefficient
364	Payment Method_UPI	1.412039
5	day_of_week	0.035604
183	Pickup Location_Vinobapuri	0.033809
204	Drop Location_Bhikaji Cama Place	0.023193
135	Pickup Location_Pragati Maidan	0.022618
223	Drop Location_Ghaziabad	0.021211
227	Drop Location_Govindpuri	0.020256
210	Drop Location_Chhatarpur	0.017910
47	Pickup Location_GTB Nagar	0.017487
35	Pickup Location_Chhatarpur	0.016738

Top Features Decreasing Cancellation (Negative Coefficients):		
	Feature	Coefficient
3	Ride Distance	-0.589528
1	Avg CTAT	-0.559496
2	Booking Value	-0.335035
0	Avg VTAT	-0.071794
363	Payment Method_Debit Card	-0.025498
91	Pickup Location_Khandsa	-0.025032
140	Pickup Location_RK Puram	-0.020510
168	Pickup Location_Subhash Nagar	-0.018339
68	Pickup Location_IMT Manesar	-0.016487
295	Drop Location_Noida Extension	-0.016398

Conclusion

- **Random Forest was the most accurate**, but it took the longest
- **XGBoosting ran the fastest**
- Tradeoffs faster model equated to lower accuracy
- Ride distance, booking value, VTAT, and CTAT are most important features

